Fig. 2 - The research record and its three layers

The paper has three layers, one for each reader; the correspondence between paper, appendix, and record is a checkable object.

The research record: what the author submits

Paper: for people; it opens with a notice that an AI appendix exists, and where. The question, what was done, what it means; any length.

Appendix: for the reader who checks. Proofs and derivations, procedures, full tables and robustness, data description; a short statement of scope and the choices made.

AI appendix: the author's statement to the machine, for the reader that evaluates. Six parts in a published form: definitions of record; the procedure, re-executable; the claims list, each claim pointed to its evidence; the authors' stance; reproduction anchors; scope and do-nots. Written with any tool, the reference generator among them, and edited freely; if none arrives, the journal's generator produces one.

Materials: data, code, outputs, or field materials, each in a declared access tier (public, restricted, described-only).

Provenance: what produced each result — code and data (empirical); the derivation (formal); seeds and runs (simulation); the audit trail from observation to interpretation (qualitative); the argument structure (verbal theory); corpus and inclusion criteria (review).

Declarations: modality · standard version · access tier · how AI was used. Working history at the author's choice.

the record

Level 2 · Correspondence (internal)
Does the paper match the appendix, and the appendix the record? Read through the AI appendix, where each claim points.
(a) Completeness: every claim the paper rests on is in the appendix, and so in the claims list.
(b) Resolution: each pointer resolves to evidence that says what the claim says.
(c) Anchors: the reproduction anchors reproduce (deterministic).
Whether the AI appendix matches the reference generator's output is recorded on the report as a fact, not a finding.
Checkable from the record alone. Fix: correct the paper or the appendix to match the record, or the record, then re-run.

then

Level 3 · Warrant (external)
Do the claims match the evidence offered, judged against what the record does not supply?
The standard for the modality: what a causal claim, a proposition, a qualitative interpretation needs (SIEVE modules: statistical, internal, construct, external validity; contribution; modality soundness).
The literature: does the contribution claim survive the nearest prior work? Output: the evidence map.
Re-execution under perturbation: do the results survive the robustness checks the standard names?
Fix: strengthen the evidence or weaken the claims.

Diff between rounds: what changed in the record between review rounds, and who changed each file, author or AI, as the declaration states. What reviewers read from the second round.

Slop is a claim that goes beyond its evidence. It is not: (a) unconventional work — strong evidence, unfamiliar approach; (b) wrong work — reasonable evidence, later overturned; (c) unimportant work — well supported, little valued.
Internal checks cannot tell a Potemkin village from a town; Level 3 is built to, and names what it cannot reach: data that never existed.

Modalities: formal built for Stage 1; empirical in Stage 2; the rest specified by protocol.

Formal theory built for Stage 1	**Empirical Stage 2**	**Simulation**	**Qualitative**	**Survey / review**	**Verbal theory**

Limit: the record cannot prove human direction. Anything checkable could be generated; the AI-use declaration and the working history rest on good faith and reputation, as authorship statements do.

What is new, taken together: a claims list as a schema; a standard AI appendix, the author's, under a published specification; paper-to-appendix-to-record correspondence checked by published code under a published standard; a published mapping from findings to actions; one indivisible certificate signed by named people, issued with the publication. Registered Reports, Code Ocean, Statcheck, and eLife each did one of these.

Journal · Author + AI · Machine · Named people · runs in Mark 1 · specified only · stretch · decision: a person only